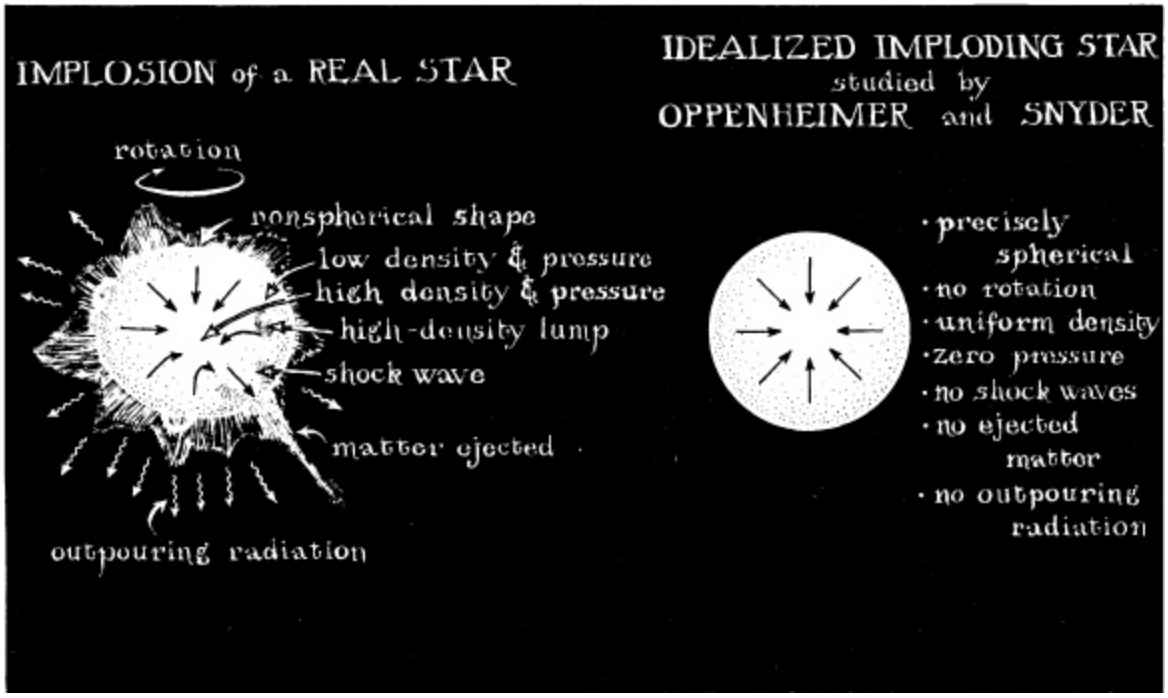
from an idealized, pressureless star; but the differences of implosion should be only modest, not great, Oppenheimer's intuition insisted.

Thus it was that Oppenheimer suggested to Snyder an idealized computational problem: Study, using the precise laws of general relativity, the implosion of a star that is idealized as precisely spherical, nonspinning, and nonradiating, a star with uniform density (the same near its surface as at its center) and with no internal pressure whatsoever; see Figure 6.3.

Even with all these idealizations—idealizations that would generate skepticism in other physicists for thirty years to come—the calculation was exceedingly difficult. Fortunately, Richard Tolman was available in Pasadena for help. Leaning heavily on Tolman and Oppenheimer for advice, Snyder worked out the equations governing the entire implosion—and in a tour de force, he managed to solve them. He now had the full details of the implosion, expressed in formulas! By scrutinizing those formulas, first from one direction and then another, physicists could read off whatever aspect of the implosion they wished—how it looks from outside the star, how it looks from inside, how it looks on the star's surface, and so forth.



6.3 *Left:* Physical phenomena in a realistic, imploding star. *Right:* The idealizations that Oppenheimer and Snyder made in order to compute stellar implosion.